Fiscal Year:	FY 2026	**Task Last Updated:**	FY 03/11/2026
PI Name:	Beheshti, Afshin Ph.D.		
Project Title:	Investigating the Impact of Spaceflight and Regolith on Mitochondrial and miRNA Function in Liver and 3D Human Tissue Models		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	beheshti@pitt.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-308-2540
Organization Name:	University of Pittsburgh		
PI Address 1:	4200 Fifth Ave		
PI Address 2:			
PI Web Page:			
City:	Pittsburgh	**State:**	PA
Zip Code:	15260	**Congressional District:**	12
Comments:	PI has moved from NASA Ames Research Center to the University of Pittsburgh, where he serves as Director of the Center for Space Biomedicine. Former email: afshin.beheshti@nasa.gov		
Project Type:	Ground	**Solicitation / Funding Source:**	2022 Space Biology NNH22ZDA001N-SBR: E.9 Space Biology Research Studies
Start Date:	01/05/2026	**End Date:**	01/04/2028
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
Contact Email:	harry.jones@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	NOTE: The Principal Investigator (PI) transitioned from Blue Marble Space Institute of Science/NASA Ames Research Center to the University of Pittsburgh in September 2024.		
COI Name (Institution):	Mason, Christopher (Weill Cornell Medicine) Schwartz, Robert (Weill Cornell Medicine) Morten, Karl (Oxford University) Galeano, Diego (National University of Asunción)		
Grant/Contract No.:	80NSSC26K0370		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project aims to investigate the effects of spaceflight and regolith exposure on mitochondrial function, miRNA regulation, and systemic health impacts in both archived rodent samples and 3D human tissue models. Our hypothesis is that spaceflight and regolith exposure increase mitochondrial DNA (mtDNA) copy number, oxidative phosphorylation (OXPHOS), and miRNA expression in the liver, leading to immune suppression, inflammation, and mitochondrial dysfunction. The project will utilize a combination of archived rodent tissues from spaceflight missions, 3D human liver and microvasculature tissue models, and omics data from NASA's Open Science Data Repository (OSDR), astronaut data, and other commercial spaceflight missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This two-year study will significantly advance our understanding of how spaceflight and regolith impact mitochondrial and immune function, with the potential to develop novel, FDA-approved countermeasures to mitigate the associated health risks for astronauts.
Task Progress:	New Project for FY2026. For previous reporting related to this investigation, please visit: https://
Bibliography Type:	Description: (Last Updated: 07/13/2026)
Articles in Peer-reviewed Journals	Yin Y, Xu R, Makarczyk MJ, Liu JJ, Liu S, Shi MZ, Carlo RD, Almeida MF, Garris A, Day S, Sanchez-Hodge R, Schisler JC, Zilberman AH, Allen NG, Kubik AJ, Blaber EA, Olali AZ, Xie W, Wallace DC, Mason CE, Alexander PG, Intini G, Beheshti A, Lin H. "Kaempferol attenuates spaceflight-associated knee cartilage degradation by targeting NOX4-mediated mitochondrial dysfunction." Adv Sci (Weinh). 2026 Jul 9;e76477. https://doi.org/10.1002/advs.76477 ; PMID: 42423501; PMCID: PMC13348333 , Jul-2026
Books/Book Chapters	Enguita FJ, Leitão AL, Beheshti A. "A systems biology approach to discovery and validation of MicroRNA signature health risks associated with the space environment." in "Methods Mol Biol. Vol. 3000. " Ed. M Bizzarri. New York, NY: Humana, 2026. p. 123-43. https://doi.org/10.1007/978-1-0716-5174-2_6 ; PMID: 42151676, May-2026